THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

DEC, 18, 2024 TIME: 2:30 pm - 5:30 pm

Answer ALL questions. Section A carries 30 marks and Section B carries 70 marks.

Use the same answer book for both Section A and B.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (Ten One-choice questions with 3 marks each.
There will not be any deduction for wrong answers.)
You may need to use Appendix A at the end of the exam paper for your calculations.

1. When to prepare 5 ppm $NaHCO_3$ in 1 L of water solution, how much $NaHCO_3$ is needed?
(A) 5 mg; (B) 5 g; (C) 84 mg; (D) 84 g.

2. Given $HOCl \rightleftharpoons OCl^- + H^+$ with the equilibrium constant of $K= 10^{-7.6}$, what is the acid in the equilibrium and what is the percentage of [HOCl] over the total chloride concentration at pH 7.6, respectively?
(A) H^+, 50%; (B) H^+, 90%; (C) HOCl, 50%; (D) HOCl, 90%

3. In a carbonate system at standard temperature and pressure, what is the main species at pH 13?
(A) H_2CO_3; (B) HCO_3^-; (C) CO_3^{2-}; (D) H_3O^+

4. Which of the following statements regarding water is **NOT** true?
(A) A dipole is a molecule or covalent bond that has a separation of charges.
(B) Surface water is any body of water above ground.
(C) Temperature could be one of physical pollutants in water due to the effect on aquatic life.
(D) Nutrients could be one of biological pollutants in water due to the effect on algal bloom.

5. For a BOD_5 determinate of wastewater, a 10 mL wastewater mixed with enough water to fill a 200 mL bottle has an initial dissolved oxygen of 8.0 mg/L. However, after 5-days incubation at 20 °C, the dissolved oxygen decreased to 5.0 mg/L. The BOD_5 of the sewage would be
(A) 100 mg/L; (B) 300 mg/L; (C) 600 mg/L; (D) Cannot be determined.

6. Which of the following statements is **True** in water treatment technologies?
(A) In sedimentation, the settling velocity of particles is determined by the gravitational force;
(B) Coagulation-flocculation is a physical water treatment technique typically applied prior to sedimentation and filtration;
(C) The objectives of disinfection are the kill pathogens in water and prevent pathogen regrowth in water;
(D) Alkalinity is the ability of a water to consume OH^-.

7. There is a small city of 500,000 send 0.6 m^3 of sewage /person /day to the wastewater treatment plant. A circular primary clarifier would be designed to have an average detention time of 0.96 hours and an average overflow rate of 100 m/d. What would be the dimension (the clarifier surface area, clarifier diameter and clarifier depth) of this clarifier?
(A) 3000 m^2, 31 m, 96 m.
(B) 3000 m^2, 62 m, 4.0 m.
(C) 375 m^2, 22 m, 3.3 m.
(D) 375 m^2, 22 m, 96 m.

8. Which of the following statements is **NOT** true for activated sludge process?
(A) It is an anaerobic process.
(B) It is a returning of sludge process;
(C) "Activated" is referring to return of active biomass from secondary tank to primary tank.
(D) Food to micro-organism ratio is an important design and operation parameter indicating the relationship between the organic loading and the number of micro-organisms.

9. Which of the following pollutants is not one of the secondary air pollutants?
(A) Sulfur trioxide (SO_3); (B) Nitrogen dioxide (NO_2); (C) Ozone (O_3); (D) Nitric acid (HNO)

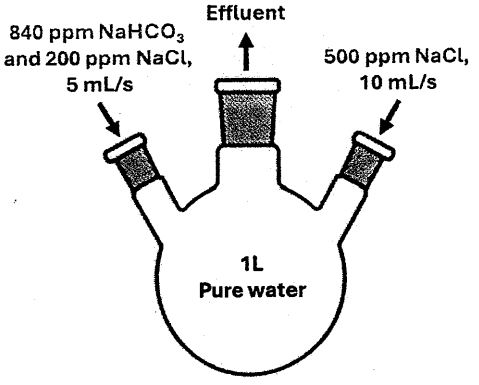
10. Which of the following is not an ozone destructor?
(A) Halons (CF_3Br; CF_2ClBr);
(B) Chlorofluorocarbons (CFCs);
(C) Perfluorocarbons (PFCs);
(D) Hydrochlorofluorocarbons (HCFCs)

Section B. (Four questions with unequal marks)

You may need to use Appendix A at the end of the exam paper for your calculations.

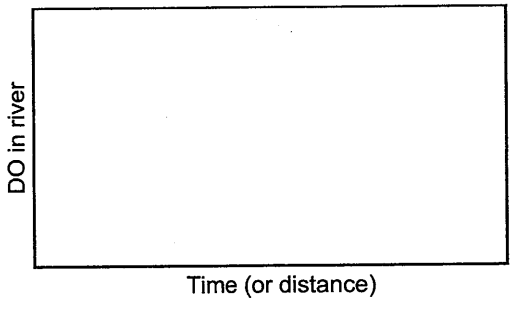
11. Consider an open three-neck flask containing 1 L of pure water (see diagram below). A water solution containing 840 ppm $NaHCO_3$ and 200 ppm NaCl is steadily poured into the flask with a flow rate of 5 mL/s from the left neck. Another water solution of 500 ppm NaCl is steadily input from the right neck with a flow rate of 10 mL/s. From the middle neck, an effluent is steadily pumped out of the flask. Assuming a steady-state system, the solution in the flask is completely mixed at standard temperature and pressure with 300 ppm CO_2 in the air, no water evaporation, and no decay/generation reaction in the system.

(a) Perform the mass balance for **water** and **NaCl** in the flask and find the steady-state concentration of NaCl in the effluent. **(4 marks)**
(b) What are the carbonate species in the flask? **(2 marks)**
(c) Perform the charge balance in the flask. **(2 marks)**
(d) What is the initial pH of the pure water in the flask? **(4 marks)**
(e) What is the pH of the solution containing 840 ppm $NaHCO_3$ (Assuming NaCl does not affect pH)? **(3 marks)**



12. A continuous discharge of hospital wastewater mixes with a river, the BOD is 10 mg/L, and the DO is 8.1 mg/L. The river and the wastewater mixture have a temperature of 20 °C, the saturated value of DO of water (DO_s) at this condition is 9.1 mg/L, a deoxygenation constant k_d of 0.20/day, an average flow speed of 0.25 m/s, and an average depth of 4.0 m.
(a) Please find the time and distance downstream at which the oxygen deficit is at a maximum (Note: the reaeration constant $k_r = \frac{3.9u^{1/2}}{H^{3/2}}$). (6 marks)

(b) Draw out the oxygen sag, deoxygenation and reaeration curves; briefly introduce these curves (< **150 words**). (9 marks)



13. (a) Rapid growth of microelectronics industry leads to a significant increase in the generation of microelectronic wastewater containing complex pollutants. This massive volume of discharged semiconductor wastewater containing ions (e.g., F^-, Cl^-, SO_4^{2-} and PO_4^{3-}), macromolecule organics (e.g., tetramethylammonium hydroxide and isopropyl alcohol), inorganic particles (e.g., SiO_2, Al_2O_3 and CeO_2), represents a significant challenge in environment. Recovery those materials is crucial for both environmental and economic reasons. Briefly design an appropriate process (different membrane technologies can be included) for F^- and SiO_2 sustainable recovery, explain your design purposes and provide your evaluation (**< 200 words including explanation of your design purposes, up to 1 Figure is allowed. Note: $SiO_2 + 4H^+ + 6F^- \rightarrow SiF_6^{2-} + 2H_2O$) and $SiF_6^{2-} + 2Na^+ \rightarrow Na_2SiF_6\downarrow$**). **(15 marks)**

(b) Membrane bioreactor (MBR) technology demonstrated highly efficient in sludge treatment. To further enhance the separation efficiency and achieve material recovery from hospital waste streams, novel concepts of MBRs and their applications are rapidly evolved. Please classify the MBRs in your mind and provide the reasons **(<100 words, Pictures are allowed to help the explanations)**. Please point out the main advantages and disadvantages of MBR technology in sludge treatment. **(10 marks)**

14. Compare to mobile source, power plants (especially coal fired power plants) contribute most of SO_x and significant portion of NO_x and particulates. Numerous studies have reported various control methods on the reduction of pollutants emissions.

(a) Describe the three types of control measures. **(3 marks)**

(b) What is SCR and explain how it reduces NO_x emissions. **(4 marks)**

(c) Both NO_x and SO_x are the principal precursor of acid rain. Define acid rain and outline its harmful effects (at least two). **(4 marks)**

(d) Instead of air pollutants emission, the combustion of coals and fuels also generate a great amount of CO_2. List three major effects of climate change on the ocean and provide explanations. **(4 marks)**

Appendix A. Information for Reference

I. Atomic weights

H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

II. Properties of water

Density = 1000 kg/m^3
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/(kg·°C)
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

III. Other constants

Ideal gas law constant = 0.082056 L·atm/(K·mol)
Henry's Law constant for CO_2: 0.0334 mol/(L·atm)
Acidity constants for H_2CO_3: $pK_1 = 6.3$; $pK_2 = 10.3$
Solubility of $CaCO_3(s)$: $pK_{so} = 8.3$
Henry's Law constant for O_2 (25 °C) = 0.00126 mol/(L·atm)
Dissociation constant of water (25 °C) $K_w = 1\times10^{-14}$
Equivalent Weight (EW) for Ca^{2+} = 20 mg/meq
EW for Mg^{2+} = 12.2 mg/meq
EW for $CaCO_3$ = 50 mg/meq

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